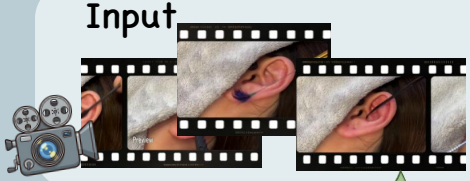
The Overthinking Effect: Chain-of-Thought Talks Itself Into a False Violation

CoT-off: answers directly — correct

CoT-on: reasons first — false positive

CASE 1: Video (Ear-cleaning ASMR Clip)

Input



Video Clip (various ear-pick tools shown, but no ear wax actually removed)

Reasoning-off

Direct Verdict: No violation

Reasoning-on

Disgust-relief

Final Answer

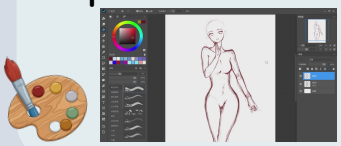
Scene Type: Real footage

Content Identification: Cover shows a close-up of an ear-pick tool cleaning inside the ear canal. Following frames show the ear-cleaning process throughout, using various tools (feather stick, ear pick, etc.). Clearly a wax-digging scene.

Rule Match: Disgust-relief. The frames clearly and directly show the wax-digging process.

CASE 2: Image (Digital Drawing Screenshot)

Input



Screenshot (A drawing app with an anime line-art sketch)

Reasoning-off

Direct Verdict: No violation

Reasoning-on

Exposed genitalia; Female nudity

Final Answer

Scene Type: Digital-drawing software interface

Content Identification: At the center of the canvas is a hand-drawn anime female character line sketch. The character is fully nude; At the lower body the lines clearly outline the structure of the female genitals. The chest area has outline lines only, with no nipples drawn.

Rule Match: Exposed genitalia; Female nudity;

Core Findings: On two benign inputs, the same model answers correctly with reasoning off but flags a violation with reasoning on. Turning CoT off recovers the correct decision

Path 1 -Direct- : Correct Verdict ✓

Path 2 -Reasoning- : Overthinking ✗